

$$1. \quad (1) \quad \frac{\partial u}{\partial x} = 2y = \frac{\partial v}{\partial y} = 2x$$

$$x=y$$

$$\frac{\partial u}{\partial y} = 2(x-1) = -\frac{\partial v}{\partial x} = -2y-3$$

$$2x-2 = -2x-3$$

$$4x = -1$$

$$x = -\frac{1}{4} \quad y = -\frac{1}{4}$$

$f(z)$ 在 $(-\frac{1}{4}, -\frac{1}{4})$ 上可导,
处处不解析

(2)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x$$

$$u = x^2 + f(y)$$

$$\frac{\partial u}{\partial y} = f'(y) = -\frac{\partial v}{\partial x} = -2y-3$$

$$f(y) = -y^2 - 3y + C$$

$$f(0) = C = 0$$

$$g(z) = x^2 - y^2 - 3y + (2xy + 3x)i$$

$$= z^2 + 3zi$$

2.

$$\oint_{|z-1|=2} \frac{\sin z}{(z-1)^6} dz$$

$$= 2\pi i \frac{1}{5!} (\sin z)^{(5)} \Big|_{z=1}$$

$$= \frac{1}{5!} \times 2\pi i = \frac{\pi i}{60}$$

$$= \frac{2\pi i}{5!} \cos 1$$

3.

$$z = 2e^{i\theta} = (2\cos\theta) + i(2\sin\theta)$$

$$\bar{z} = 2e^{-i\theta} = (2\cos\theta) - i(2\sin\theta)$$

$$\oint_{|z+1|=2} (\bar{z} + 2021z^2 - \cos z) dz$$

$$= \oint_{|z+1|=2} \bar{z} dz = (2\pi i) \cdot \frac{1}{2}$$

$$= \int_0^{2\pi} (2e^{-i\theta} + i)(2ie^{i\theta}) d\theta$$

$$= 8\pi i$$

$$1. \quad \int_0^{+\infty} \frac{x^2}{(x^2+9)(x^2+16)} dx$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x^2}{(x^2+9)(x^2+16)} dx$$

$$= \pi i \left[\text{Res} \left[\frac{x^2}{(x+3i)(x^2+16)}, 3i \right] + \text{Res} \left[\frac{x^2}{(x+9)(x+4i)}, 4i \right] \right]$$

$$= \pi i \left[\frac{-9}{6i \times 7} + \frac{-16}{-7 \times 8i} \right] = -\frac{\pi}{14}$$

2.

$$\ln(1+\sqrt{3}i) + \left(\frac{1-i}{1+i}\right)^{2021}$$

$$= \ln 2 + \left(\frac{\pi}{3} + 2k\pi\right)i - i$$

$$= \ln 2 + \left(\frac{\pi}{3} - 1 + 2k\pi\right)i$$

$$3. \quad f(z) = \frac{1}{1+z+1} + e \cdot e^{-z-1}$$

$$= \sum_{n=0}^{\infty} (-z-1)^n + e \cdot \sum_{n=0}^{\infty} \frac{(-z-1)^n}{n!}$$

$$= \sum_{n=0}^{\infty} (-1)^n (z+1)^n \left(1 + \frac{e}{n!}\right)$$

$$\left| \frac{1 + \frac{e}{(n+1)!}}{1 + \frac{e}{n!}} \right| = 1 \quad R=1$$

$$f(z) = \sum_{n=0}^{\infty} (-1)^n \cdot 2^n \cdot \left(1 + \frac{e}{n!}\right)$$

不收敛

$$\therefore \text{收敛圆 } |z+1| < 1 \quad R=1$$

三.

$$z=0, \text{ 三级极点}$$

$$z=\frac{1}{2}, \text{ 一级极点}$$

$$z=\infty, \text{ 可去奇点}$$

$$|z| < \frac{1}{2}, \quad f(z) = \frac{-1}{z^3} \cdot \frac{1}{1-2z} = \sum_{n=0}^{\infty} \frac{-1}{z^3} \cdot (2z)^n$$

$$= \sum_{n=0}^{\infty} -2^n \cdot z^{n-3}$$

$$|z| > \frac{1}{2}, \quad f(z) = \frac{1}{2z^4} \cdot \frac{1}{1-\frac{1}{2z}}$$

$$= \sum_{n=0}^{\infty} \frac{1}{2z^4} \cdot \left(\frac{1}{2z}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2^{n+4} z^{n+4}}$$

四.

基础解系为 $\frac{1}{4\pi rmm_0}$

$$G = \frac{1}{4\pi rmm_0} - \frac{1}{4\pi rmm_1} - \frac{1}{4\pi rmm_2} + \frac{1}{4\pi rmm_3}$$

$$= \frac{1}{4\pi} \left(\frac{1}{rmm_0} - \frac{1}{rmm_1} - \frac{1}{rmm_2} + \frac{1}{rmm_3} \right)$$

$$u(m) = -\oint \frac{\partial \phi}{\partial n} ds$$

$$= \int_{-\infty}^{-100} \int_{-\infty}^{+\infty} \frac{\partial \phi}{\partial z} dy \int_0^{+\infty} \frac{\partial \phi}{\partial x} dy dz$$

五.

1. $\frac{\partial u}{\partial x} = 5y + f(x)$

$$u = 5xy + \int f(x) + g(y)$$

$$\int f(x) + g(y) = y$$

$$5x + \int f(x) + g(y) = \cos x$$

$$\int f(x) + g(y) = 1$$

$$\int f(x) + 5x + g(y) + \int f(x) + g(y) = y + \cos x$$

$$\int f(x) + g(y) = y + \cos x - 5x - 1$$

$$u = 5xy + y + \cos x - 5x - 1$$

2.

$$a = 0$$

$$J_2 - J_0(x)$$

$$J_2(x) - J_0(x) = b J_1'(x)$$

$$J_0'(x) = -J_1(x)$$

$$2J_1'(x) = J_{\mu-1}(x) - J_{\mu+1}(x)$$

$$J_2 - J_0 = -2J_1'(x)$$

$$\frac{d}{dx} \left[\frac{J_{\mu}(x)}{x^{\mu}} \right] = -\frac{J_{\mu+1}(x)}{x^{\mu}}$$

$$J_0''(x) = -J_1'(x)$$

$$b = 2, c = 0$$

3. $u_{xx} - u_{xy} - 12u_{yy} = 0$

$$\frac{u_{xx}}{u_{yy}} - \frac{u_x}{u_y} - 12 = 0$$

$$\left(\frac{dy}{dx} \right)^2 + \frac{dy}{dx} - 12 = 0$$

$$\left(\frac{dy}{dx} + 4 \right) \left(\frac{dy}{dx} - 3 \right) = 0$$

$$\frac{dy}{dx} = 3 \text{ 或 } -4$$

$$y = 3x + C_1 \text{ 或 } y = -4x + C_2$$

$$\xi = 3x + y, \eta = 4x + y$$

$$u = f_1(\xi)$$

$$\frac{\partial u}{\partial x} = -3 \frac{\partial u}{\partial \xi} + 4 \frac{\partial u}{\partial \eta}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$$

$$\frac{\partial^2 u}{\partial x^2} = 9 \frac{\partial^2 u}{\partial \xi^2} - 12 \frac{\partial^2 u}{\partial \xi \partial \eta} + 16 \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = -3 \left(\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \xi \partial \eta} \right)$$

$$+ 4 \left(\frac{\partial^2 u}{\partial \eta^2} + \frac{\partial^2 u}{\partial \xi \partial \eta} \right)$$

$$= -3 \frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \xi \partial \eta} + 4 \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$$

$$u = f_1(\xi) + f_2(\eta)$$

$$= f_1(-3x + y) + f_2(4x + y)$$

$$f_1(-3x) + f_2(4x) = 3x^2$$

$$f_1'(-3x) + f_2'(4x) = 0$$

$$= \frac{1}{3} f_1(-3x) + \frac{1}{4} f_2(4x) = C$$

$$\frac{1}{3} f_1(-3x) = 3x^2 - 4C$$

$$\frac{1}{4} f_2(4x) = 3x^2 + 3C$$

$$f_1(-3x) = \frac{9}{1} x^2 - \frac{12C}{1}$$

$$f_2(4x) = \frac{12x^2}{1} + \frac{12C}{1}$$

$$f_1(x) = \frac{1}{7}x^2 + \frac{12}{7}C$$

$$f_2(x) = \frac{3}{28}x^2 + \frac{12}{7}C$$

$$f_1(-3x+y) = \frac{1}{7}(-3x+y)^2 + \frac{12}{7}C$$

$$f_2(4x+y) = \frac{3}{28}(4x+y)^2 + \frac{12}{7}C$$

$$u(x, y) = \frac{1}{7}(-3x+y)^2 + \frac{3}{28}(4x+y)^2$$

$$1. \begin{cases} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} = 0. \\ u|_{\rho=R} = \sin 2020\theta \end{cases}$$

$$\hat{u} = R(\rho)\phi(\theta)$$

$$\frac{\rho^2 R''(\rho)}{R(\rho)} + \frac{\rho R'(\rho)}{R(\rho)} = -\frac{\phi''(\theta)}{\phi(\theta)} = \lambda$$

$$\begin{cases} \phi''(\theta) + \lambda \phi(\theta) = 0. \\ \phi(0) = \phi(2\pi) \\ \phi'(0) = \phi'(2\pi) \end{cases}$$

$$\lambda < 0. \quad \phi(\theta) = A e^{\sqrt{\lambda}\theta} + B e^{-\sqrt{\lambda}\theta}$$

$$A = B = 0$$

$$\lambda = 0 \quad \phi(\theta) = A\theta + B$$

$$\phi(\theta) = B.$$

$$\lambda > 0. \quad \phi(\theta) = A \cos \beta\theta + B \sin \beta\theta$$

$$\lambda = \beta^2$$

$$\beta = n, \quad \lambda = n^2$$

$$\phi(\theta) = A \cos n\theta + B \sin n\theta$$

$$\rho^2 R''(\rho) + \rho R'(\rho) - \lambda R(\rho) = 0.$$

$$\lambda = 0. \quad D^2 R(\rho) = 0. \quad R(\rho) = A\rho + B$$

$$R(\rho) = A \ln \rho + B$$

$$\lambda > 0. \quad D^2 R(\rho) - n^2 R(\rho) = 0.$$

$$R(\rho) = A' e^{\sqrt{\lambda}\rho} + B' e^{-\sqrt{\lambda}\rho}$$

$$B(\rho) = A' \rho^n + B' \rho^{-n}$$

$$|R(\infty)| < \infty.$$

$$A = 0. \quad B' = 0.$$

$$R(\rho) = C_n \rho^n$$

$$u(\rho, \theta) = A + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) \rho^n$$

$$A + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) R^n = \sin 2020\theta$$

$$A = A_n = 0.$$

$$\sum_{n=1}^{\infty} B_n \sin 2020\theta R^{2020} = \sin 2020\theta$$

$$B_n = \frac{1}{R^{2020} \pi} \int_0^{2\pi} \sin^2 2020\theta d\theta$$

$$= \frac{1}{R^{2020}}$$

$$u(\rho, \theta) = \sum_{n=1}^{\infty} \frac{1}{R^{2020}} \sin n\theta \rho^n$$

$$2. \begin{cases} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - 2 & 0 < x < 1, t > 0. \\ u|_{x=0} = -1 \quad u|_{x=1} = 0. & 0 \leq x \leq 1, t > 0 \\ u|_{t=0} = x^2. & 0 \leq x \leq 1 \end{cases}$$

$$u(x, t) = v(x, t) + w(x)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial t} \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 w}{\partial x^2}$$

$$\begin{cases} \frac{\partial^2 w}{\partial x^2} - 2 = 0 \\ w(0) = -1 \\ w(1) = 0 \end{cases}$$

$$w(0) = -1$$

$$w(1) = 0$$

$$w(x) = x^2 - 1$$

$$\begin{cases} \frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} \\ v|_{x=0} = 0 \quad v|_{x=1} = 0 \\ v|_{t=0} = 1 \end{cases}$$

$$v(x, t) = X(x)T(t)$$

$$X(x)T'(t) = X''(x)T(t)$$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -\lambda$$

$$X''(x) + \lambda X(x) = 0.$$

$$\lambda < 0. \quad X(x) = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}$$

$$A = B = 0.$$

$$\lambda = 0. \quad X(x) = Ax + B$$

$$A = B = 0.$$

$$\lambda > 0. \quad X(x) = A \cos \beta x + B \sin \beta x$$

$$\lambda = \beta^2$$

$$A = 0.$$

$$\sin \beta = 0. \quad \beta = n\pi$$

$$\lambda = n^2 \pi^2$$

$$X(x) = C_1 \sin n\pi x$$

$$T(t) = A_n \cos n\pi t + B_n \sin n\pi t$$

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos n\pi t + B_n \sin n\pi t) \sin n\pi x$$

$$\neq \sum_{n=1}^{\infty} A_n \sin n\pi x = 1$$

$$A_n = 2 \int_0^1 \sin n\pi x \, dx$$

$$= \frac{2}{n\pi} (1 - \cos n\pi)$$

$$T'(t) + \lambda^2 T(t) = 0$$

$$T(t) = e^{-n^2 \pi^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-n^2 \pi^2 t} \sin n\pi x$$

$$\sum_{n=1}^{\infty} C_n \sin n\pi x = 1$$

$$C_n = 2 \int_0^1 \sin n\pi x \, dx$$

$$= \frac{2}{n\pi} (1 - \cos n\pi)$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) e^{-n^2 \pi^2 t} \sin n\pi x$$

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) e^{-n^2 \pi^2 t} \sin n\pi x + x^2 - 1$$